

SUDHANSHU GUPTA (HE/HIM)

Silicon Performance Architect, Meta Reality Labs

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EDUCATION

University of Rochester, NY, USA

August 2019 - August 2025

Ph.D., Computer Science

GPA: 3.75/4.00

Dissertation: Managing Data Movement Bottlenecks in Accelerator-Rich Systems

Advisor: Prof. Sandhya Dwarkadas

University of Rochester, NY, USA

August 2019 - May 2021

M.S., Computer Science

GPA: 3.82/4.00

NIIT University, RJ, India

July 2015 - July 2019

B.Tech., Computer Science and Engineering

CGPA: 8.87/10.00

EXPERIENCE

Meta Platforms

August 2025 - Present

Silicon Performance Architect, Meta Reality Labs

Sunnyvale, CA, USA

- Building the next generation of SoCs for Meta's wearables.
- SoC performance analysis, focusing on data movement between compute units and the main memory, using industry-standard tools.
- Workload-aware hardware/software co-design to optimize for performance and power efficiency.
- Design and development of performance and power models in C++ and SystemC.
- Workflow automation and quantitative analysis using Python.

University of Rochester

August 2019 - August 2025

Graduate Research Assistant, advised by Prof. Sandhya Dwarkadas

Rochester, NY, USA

- Investigated data movement issues in current and future processors, focusing on:
 1. Accelerator scheduling to reduce off-chip data movement and provide quality-of-service.
 2. Interconnect and memory controller design for PIM-enabled processors.
 3. Characterizing the impact of emerging heterogeneous memory on accelerator performance.

AMD Research

May 2022 - December 2022

Intern/Co-op, Core and SoC Architecture Team

Austin, TX, USA

- Contributed to the interconnect/memory controller interface for future architectures.
- Explored memory controller scheduling for PIM-enabled architectures.

Indian Institute of Technology, Delhi

January 2019 - July 2019

Research Intern, supervised by Prof. Smruti Ranjan Sarangi

New Delhi, DL, India

- Designed a benchmark suite sporting state-of-the-art CNN-based autonomous driving applications.
- Extended *Tejas* simulator to characterize compute and memory deficiencies in modern out-of-order CPUs, demonstrating CPUs outperforming GPUs in multi-application scenarios.

Indian Institute of Technology, Kanpur

May 2018 - July 2018

Research Intern (SURGE), supervised by Prof. Biswabandan Panda

Kanpur, UP, India

- Studied memory subsystem optimizations for machine learning applications running on CPUs.
- Characterized the memory performance of emerging machine learning applications using *ChampSim* simulator, demonstrating the inefficacy of state-of-the-art cache prefetchers.

PUBLICATIONS

1. **S. Gupta** and S. Dwarkadas, “Improving the Performance of Out-of-Core LLM Inference Using Heterogeneous Host Memory,” *2025 IEEE International Symposium on Workload Characterization (IISWC)*, Irvine, CA, USA, 2025, pp. 325-338, doi: 10.1109/IISWC66894.2025.00035.
2. **S. Gupta**, N. Madan, S. Puthoor, N. Jayasena and S. Dwarkadas, “Concurrent PIM and Load/Store Servicing in PIM-Enabled Memory,” *2025 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, Ghent, Belgium, 2025, pp. 320-334, doi: 10.1109/ISPASS64960.2025.00037.
3. **S. Gupta** and S. Dwarkadas, “RELIEF: Relieving Memory Pressure In SoCs Via Data Movement-Aware Accelerator Scheduling,” *2024 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Edinburgh, United Kingdom, 2024, pp. 1063-1079, doi: 10.1109/HPCA57654.2024.00084.
4. D. Moolchandani, **S. Gupta**, A. Kumar and S. R. Sarangi, “Performance Prediction for Multi-Application Concurrency on GPUs,” *2020 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, Boston, MA, USA, 2020, pp. 306-315, doi: 10.1109/ISPASS48437.2020.00050.

TECHNICAL SKILLS

- **Computer languages:** C, C++, CUDA, Python, Java, Bash, OpenMP, MPI, Verilog
- **Architectural simulators:** gem5, gem5-SALAM, gem5-Aladdin, GPGPU-Sim, ChampSim, Tejas
- **Tools:** Intel Pin, Perf, Git, Valgrind, GDB, Nvidia NSight

PROFESSIONAL SERVICE

- **Program Committee:** IISWC 2026, ISCA 2026
- **Artifact Evaluation Committee:** ISCA 2024, ISCA 2023

TEACHING EXPERIENCE

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| University of Rochester | Rochester, NY, USA |
| • Teaching Assistant for CSC 252: Computer Organization (Spring 2020, Fall 2020, Spring 2021) | |
| NIIT University | Neemrana, RJ, India |
| • Teaching Assistant for CS 382: Programming Tools (Fall 2018) | |
| Indian Institute of Technology, Kanpur | Kanpur, UP, India |
| • Teaching Assistant Computer Architecture Summer School 2018 (Summer 2018) | |

COMMUNITY

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| University of Rochester | Rochester, NY, USA |
| • Serving in the Graduate Labor Union Organizing Committee. | |
| • Computer Science Events Committee co-chair. | |
| NIIT University | Neemrana, RJ, India |
| • Technical Lead at FOSS Society, an umbrella society housing the Firefox Community and Linux Users Group. | |
| • Founder and Club Lead at Linux Users Group, a group of enthusiastic users and learners of Linux. | |
| • Member of President’s Student Advisory Board, advising the president on student affairs. | |